

YUCHEN YANG

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Bio

Dr. Yuchen Yang is an Assistant Professor in the College of Information Sciences and Technology at Penn State. Her research focuses on **ML systems security** and **trustworthy AI**. Her work has appeared at IEEE S&P, ICML, ACM CCS, USENIX Security, NDSS, EMNLP, and ECCV, and has been featured by MIT Technology Review and IEEE Spectrum. She is a recipient of the NSF NAIRR Pilot Awards in 2025 and 2026 and the Penn State SSRI Social AI Award in 2026.

ACADEMIC AND RESEARCH APPOINTMENTS

Assistant Professor The Pennsylvania State University , PA, United States	2025.07 - Current
Graduate Research Assistant for Dr. Yinzhi Cao The Johns Hopkins University , MD, United States	2020.03 - 2025.05
Research Intern for Dr. Shao-yuan Lo , Dr. Kwonjoon Lee , and Dr. Behzad Dariush Honda Research Institute , CA, United States	2023.10 - 2024.02

EDUCATION

Ph.D. The Johns Hopkins University <i>Computer Science</i> Advisor: Dr. Yinzhi Cao	2021 – 2025 MD, United States
M.S. The Johns Hopkins University <i>Security Informatics</i> Advisor: Dr. Yinzhi Cao	2019 – 2021 MD, United States
B.E. Shandong University <i>Software Engineering</i>	2015 – 2019 Shandong, China

AWARDS AND GRANTS

AWARDED

Recipient of Penn State SSRI (Social Science Research Institute) Social AI Award, " <i>Persuasion-Aware Explanations for Safer AI-Mediated Decisions</i> ", PI, \$19,680	2026.03
Recipient of NSF NAIRR Pilot Award, " <i>Certifying Internal Memory in Efficient LLMs: Robustness and Integrity of the KV-Cache</i> ", PI, 15,000 GPU-hours on PSC Bridges-2	2026.01
Recipient of NSF NAIRR Pilot Award, " <i>Detect Scams in the Wild: Towards Realistic Benchmarks and Victim-Centered Evaluation</i> ", PI, \$5,000 OpenAI credits and 6,200 GPU-hours on PSC Bridges-2	2026.01
Recipient of USENIX Security Travel Grant	2025.07
Selected as ICLR Notable Reviewer	2025.05
Academic Scholarship , Shandong University	2018.09

PUBLICATIONS

*Underlined authors are students under my supervision; * indicates equal contribution.*

PREPRINTS/UNDER REVIEW

Lost in Compaction: Evaluating Side-Constraint Loss under Context Compaction

Zhiqi Wang, Yichi Zhang, Dongwon Lee, **Yuchen Yang**

The Refusal Signal Survives Quantization but Goes Unconsumed

Deekshith Reddy Bhoomireddy, Rishit Rai, Zhiqi Wang, Yichi Zhang, **Yuchen Yang**

Rethinking Quality in Large Language Model Routing: Security-Conditioned Model-and-Budget Selection

Zhenxiang Huang, Yichi Zhang, Yuyang Zhang, **Yuchen Yang**

ScamBridge: Bridging Language Model and Human in Scam Detection through Psychological Techniques

Zhiqi Wang, Yichi Zhang, Xiaozhi Yang, Dongwon Lee, **Yuchen Yang**

A Plausible Explanation Is Enough: Short-Circuiting Visual Verification in Multimodal Judges

Anglin Cai, Xiaozhi Yang, Yurun Lin, Zhiqi Wang, Dongwon Lee, **Yuchen Yang**

Steering to Say No: Configurable Refusal via Activation Steering in Vision Language Models

Jiaxi Yang, Shicheng Liu, **Yuchen Yang**, Dongwon Lee

PEER-REVIEWED PAPERS

2027 **Secure Speculative Decoding for Large Language Models**

Yichi Zhang, Zhiqi Wang, Neil Gong, **Yuchen Yang**

To appear in the Proceedings of the IEEE Symposium on Security and Privacy (S&P), 2027

First Round Acceptance Rate: 15.8%

2026 **HijackKV: New Threat in Position-Independent KV Cache Reuse**

Yichi Zhang, Zhiqi Wang, Huan Zhang, **Yuchen Yang**

In the proceedings of USENIX Security Symposium (USENIX), 2026

Acceptance Rate: 11.9%

Earned all Artifact Badges: Artifacts Available, Artifacts Functional, Results Reproduced

AOR-Bench: Do Large Audio Language Models Over-Refuse Pseudo-Harmful Queries?

Jiaxi Yang, Chaewan Chun, Jason S. Lucas, **Yuchen Yang**, Dongwon Lee

To appear in the proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP Main Conference), 2026

Acceptance Rate: 15.4%

Rethinking Forgery Attacks on Semantic Watermarks in Black-Box Settings: A Geometric Distortion Perspective

Cheng-Yi Lee, Yichi Zhang, **Yuchen Yang**, Chun-Shien Lu, Jun-Cheng Chen

In the Proceedings of the International Conference on Machine Learning (ICML), 2026

Jailbreaking Safeguarded Text-to-Image Models via Large Language Models

Zhengyuan Jiang, Yuepeng Hu, **Yuchen Yang**, Yinzhi Cao, Neil Gong

In the Findings of the European Chapter of the Association for Computational Linguistics (EACL), 2026

AdaProb: Efficient Machine Unlearning via Adaptive Probability

Zihao Zhao, **Yuchen Yang**, Anjalie Field, Yinzhi Cao

In the proceedings of International Conference on Learning Representations (ICLR) Workshop DATA-FM, 2026

2025 **CertPHash: Towards Certified Perceptual Hashing via Robust Training**

Yuchen Yang, Qichang Liu, Christopher Brix, Huan Zhang, Yinzhi Cao

In the proceedings of USENIX Security Symposium (USENIX), 2025

- 2024 **Follow the Rules: Reasoning for Video Anomaly Detection with Large Language Models**
Yuchen Yang, Kwonjoon Lee, Behzad Dariush, Yinzhi Cao, Shao-Yuan Lo
In the Proceedings of the European Conference on Computer Vision (ECCV), 2024
- SafeGen: Mitigating Sexually Explicit Content Generation in Text-to-Image Models**
Xinfeng Li*, **Yuchen Yang***, Jiangyi Deng*, Chen Yan, Yanjiao Chen, Xiaoyu Ji, Wenyuan Xu
In the Proceedings of the ACM Conference on Computer and Communications Security (CCS), 2024
- RippleCOT: Amplifying Ripple Effect of Knowledge Editing in Language Models via Chain-of-Thought In-Context Learning**
Zihao Zhao, **Yuchen Yang**, Yijiang Li, Yinzhi Cao
In the Findings of the Association for Computational Linguistics: Empirical Methods in Natural Language Processing (EMNLP), 2024
- SneakyPrompt: Jailbreaking Text-to-image Generative Models**
Yuchen Yang, Bo Hui, Haolin Yuan, Neil Gong, Yinzhi Cao
In the Proceedings of the IEEE Symposium on Security and Privacy (S&P), 2024
[Media Coverage \(Selected\): MIT Technology Review](#) 
- 2023 **PrivateFL: Accurate, Differentially Private Federated Learning via Personalized Data Transformation**
Yuchen Yang*, Bo Hui*, Haolin Yuan*, Neil Gong, Yinzhi Cao
In the proceedings of USENIX Security Symposium (USENIX), 2023
[Earned all Artifact Badges: Artifacts Available, Artifacts Functional, Results Reproduced](#)
- Fortifying Federated Learning against Membership Inference Attacks via Client-level Input Perturbation**
Yuchen Yang, Haolin Yuan, Bo Hui, Neil Gong, Neil Fendley, Philippe Burlina, Yinzhi Cao.
In the proceedings of IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023
- 2022 **Addressing Heterogeneity in Federated Learning via Distributional Transformation**
Haolin Yuan*, Bo Hui*, **Yuchen Yang***, Philippe Burlina, Neil Gong, Yinzhi Cao
In the proceedings of the European Conference on Computer Vision (ECCV), 2022
- 2021 **Practical Blind Membership Inference Attack via Differential Comparisons**
Bo Hui*, **Yuchen Yang***, Haolin Yuan*, Philippe Burlina, Neil Gong, Yinzhi Cao
In the proceedings of Network & Distributed System Security Symposium (NDSS), 2021

RESEARCH ADVISING

PH.D. STUDENTS

Yuyang Zhang, the Pennsylvania State University, advisor	2026.08 - current
Yichi Zhang, the Pennsylvania State University, advisor	2025.08 - current
Zhiqi Wang, the Pennsylvania State University, advisor	2025.08 - current

MASTER STUDENTS

Anglin Cai, master student at NWU	2026.03 - current
Zhenxiang Huang, master student at HKU	2026.01 - current
Yurun Lin, master student at Drexel University	2025.10 - 2026.06
Adila Abudurehman, master student at JHU. Now: Software engineer at Microsoft	2023.10 - 2024.02
Manoj Valeti, master student at JHU. Now: Software engineer at AWS Amazon	2023.10 - 2024.02
Amodini Vardhan, master student at JHU. Now: Security engineer at Twilio	2023.10 - 2024.02
Ashi Garg, master student at JHU. Now: Research associate at HLTCOE	2022.09 - 2023.03

UNDERGRADUATE STUDENTS

Scott Garcia, undergraduate student at Penn State, advisor for Honors Program Thesis	2026.03 - current
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Deekshith Reddy Bhoomireddy, undergraduate student at Penn State	2025.11 - current
Rishit Rai, undergraduate student at Penn State	2025.11 - current
Zihao Zhao, undergraduate student at JHU. Now Ph.D. student at JHU	2023.12 - 2024.09
Qichang Liu, undergraduate student at Tsinghua University. Now Ph.D. student at UIUC	2024.06 - 2024.09
Yichen Li, undergraduate student at Beijing University	2024.06 - 2024.09
Anning Li, undergraduate student at UESTC. Now: Master student at CMU	2023.06 - 2023.09

TEACHING EXPERIENCE

Instructor , IST597: Secure and Efficient Inference for LLMs & Agents, Penn State	Spring 2027 (Upcoming)
Instructor , IST362: Cybersecurity Analytics Studio, Penn State	Spring 2026
Instructor , IST866: Threats and Cybersecurity, Penn State	Fall 2025
Teaching Assistant , EN 601.340/440/640: Web Security, JHU	Fall 2020, 2022

SYNERGISTIC ACTIVITIES

FUNDING PANELIST

Panelist , NSF Future CoRe III	2026
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CONFERENCE/JOURNAL REVIEWING

PC , USENIX Security Symposium	2026
European Conference on Computer Systems (EuroSys)	2026
IEEE Symposium on Security and Privacy (S&P)	2026, 2027
ACM Conference on Computer and Communications Security (CCS)	2025, 2026
ACM Workshop on Adaptive and Autonomous Cyber Defense (AACD)	2024
AEC , IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)	2024
Reviewer , International Conference on Learning Representations (ICLR)	2025
International Conference on Computer Vision (ICCV)	2025
IEEE Transactions on Dependable and Secure Computing (TDSC)	2023, 2024
IEEE Transactions on Information Forensics & Security (T-IFS)	2024
External Reviewer , IEEE Symposium on Security and Privacy (S&P)	2025
USENIX Security Symposium	2023, 2024
ACM Conference on Computer and Communications Security (CCS)	2022
ACM ASIA Conference on Computer and Communications Security (ASIACCS)	2024
IEEE Computer Security Foundations Symposium (CSF)	2022, 2024
IEEE International Conference on Distributed Computing Systems (ICDCS)	2022

ORGANIZING AND CHAIRING

Session Chair , IEEE Workshop on Deep Learning Security and Privacy (DLSP)	2024
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MEDIA COVERAGE

Text-to-image AI models can be tricked into generating disturbing images, MIT Technology Review	2023.11
AI art generators can be fooled into making NSFW images, IEEE Spectrum	2023.11
Nonsense prompts trick AIs into producing NSFW images, Technology Networks	2023.11
Researchers reveal vulnerabilities in AI models, prompting concerns, Cryptopolitan	2023.11
AI generators can be tricked into making NSFW content, JHU Engineering School News	2023.11
SneakyPrompt: Revealing the vulnerabilities of text-to-image AI, The Johns Hopkins News-Letter	2023.12

PATENT

- 2025 **System and Method Using Reasoning for Video Anomaly Detection with Large Language Models**
Yuchen Yang, Kwonjoon Lee, Shao-Yuan Lo, Behzad Dariush
Under US Patent Application (No.20250259448), 2025
- 2018 **A New System for Stock Volatility Prediction by Using Privileged Support Vector Machines**
Yuchen Yang, Zheng Yan, Haoyang Li, Zhixuan Lv, Weiwei Zhao, Simiao Zhao
Under IP Australia Application (No.2018101304), 2018

INVITED TALKS

- "Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems"*, The Pennsylvania State University, **Invited talk** 2025.3
- "Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems"*, Case Western Reserve University, **Invited talk** 2025.2
- "Diagnose, Correct, Steer: Towards Functional and Trustworthy AI Systems"*, Rice University, **Invited talk** 2025.2
- "Diagnose, Correct, Steer: Towards Functional, Trustworthy AI"*, Virginia Tech, **Invited talk** 2025.2
- "Diagnose, Correct, Steer: Towards Functional, Trustworthy AI"*, Rochester Institute of Technology, **Invited talk** 2025.1
- "Diagnose, Correct, Steer: Towards Functional, Trustworthy AI"*, Monash University, **Invited talk** 2024.10
- "Zero-shot Video Anomaly Detection: Steering LLMs for Rules-based Reasoning"*, Voxel51, **Invited talk** 2024.11